Variance and Standard Deviation

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1. Variance of Daily Step Counts

Suppose the daily step counts for 5 days are:

5,000, 6,000, 7,000, 8,000, 9,000

Step 1: Find the Mean

Mean = (5,000 + 6,000 + 7,000 + 8,000 + 9,000) ÷ 5

Mean = 7,000

Step 2: Find the deviation from the mean

Steps	Deviation from Mean	Squared Deviation
5,000	-2,000	4,000,000
6,000	-1,000	1,000,000
7,000	0	0
8,000	1,000	1,000,000
9,000	2,000	4,000,000

Sum of squared deviations:

10,000,000

Step 3: Calculate Variance

Variance = Sum of squared deviations ÷ Number of observations

Variance = 10,000,000 ÷ 5

Variance = 2,000,000

Conclusion: The variance shows how much the daily step counts are spread around the mean.

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2. Standard Deviation of Monthly Mobile Data Usage

Suppose the monthly mobile data usage is:

10 GB, 12 GB, 11 GB, 13 GB, 14 GB

Step 1: Find the Mean

Mean = (10 + 12 + 11 + 13 + 14) ÷ 5

Mean = 12 GB

Step 2: Calculate squared deviations

Data Usage	Deviation	Squared Deviation
10	-2	4
12	0	0
11	-1	1
13	1	1
14	2	4

Sum = **10**

Step 3: Find Variance

Variance = 10 ÷ 5

Variance = 2

Step 4: Find Standard Deviation

Standard Deviation = √Variance

Standard Deviation = √2

Standard Deviation ≈ 1.41 GB

Interpretation

For this assignment, we can set:

- **Low Standard Deviation:** Less than 3 GB → Consistent usage
- **High Standard Deviation:** 3 GB or more → Varied usage

Since **1.41 GB < 3 GB**:

Mobile data usage is Consistent.

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3. Comparing Variation in Two Tests

Suppose the marks are:

Coding Test

60, 65, 70, 75, 80

Mean = **70**

Variance:

Marks	Deviation	Squared Deviation
60	-10	100
65	-5	25
70	0	0
75	5	25
80	10	100

Sum = 250

Variance = 250 ÷ 5 = 50

Standard Deviation = √50 ≈ 7.07

Music Quiz

40, 55, 70, 85, 100

Mean = **70**

Marks	Deviation	Squared Deviation
40	-30	900
55	-15	225
70	0	0
85	15	225
100	30	900

Sum = 2,250

Variance = 2,250 ÷ 5 = 450

Standard Deviation = √450 ≈ 21.21

Comparison

Test	Variance	Standard Deviation
Coding Test	50	7.07
Music Quiz	450	21.21

Conclusion

The **Music Quiz has more variation in scores** because its variance and standard deviation are much higher than those of the Coding Test.

This means the students' marks in the Music Quiz are **more spread out**, while the Coding Test scores are **more consistent**.

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4. Standard Deviation for Empty or Single-Value Data

In statistics, standard deviation measures how much values vary from their mean.

Case 1: Empty Dataset

If there are **no values**, there is no variation to measure.

Standard Deviation = 0

Case 2: One Value

For example:

Data = 50

Mean = 50

Deviation = 50 - 50 = 0

Therefore:

Variance = 0

Standard Deviation = $\sqrt{0} = 0$

Conclusion

Data Situation	Standard Deviation
Empty dataset	0
Only one value	0

For this assignment, both cases are considered to have **0 standard deviation** because there is no measurable variation in the data.

Final Conclusion

Variance measures the average squared difference of observations from their mean. **Standard deviation** is the square root of variance and shows how much the data is spread around the mean.

- **Low standard deviation → Data is more consistent**
- **High standard deviation → Data has more variation**

i dont need code

Bilkul bro 👍 **No code at all.** Ye pure **Statistics/Maths theory + calculations** format mein hai, directly assignment mein submit kar sakte ho.

STATISTICS ASSIGNMENT

Topic: Variance and Standard Deviation

Name: Bhavya Jani
Course: Data Science and Data Analytics
Subject: Statistics

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1. Variance of Daily Step Counts

Suppose the daily step counts are:

5,000, 6,000, 7,000, 8,000, 9,000

Step 1: Calculate Mean

Mean = (5,000 + 6,000 + 7,000 + 8,000 + 9,000) ÷ 5

Mean = 7,000

Step 2: Calculate Squared Deviations

Daily Steps	Deviation from Mean	Squared Deviation
5,000	-2,000	4,000,000
6,000	-1,000	1,000,000
7,000	0	0

8,000	1,000	1,000,000
9,000	2,000	4,000,000

Sum of squared deviations = **10,000,000**

Step 3: Calculate Variance

Variance = 10,000,000 ÷ 5

Variance = 2,000,000

Conclusion: The variance shows the amount of variation in the person's daily step counts.

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2. Standard Deviation of Monthly Mobile Data Usage

Suppose monthly mobile data usage is:

10 GB, 12 GB, 11 GB, 13 GB, 14 GB

Step 1: Calculate Mean

Mean = (10 + 12 + 11 + 13 + 14) ÷ 5

Mean = 12 GB

Step 2: Calculate Squared Deviations

Data Usage	Deviation	Squared Deviation
10 GB	-2	4
12 GB	0	0
11 GB	-1	1
13 GB	1	1
14 GB	2	4

Sum of squared deviations = **10**

Step 3: Calculate Variance

Variance = 10 ÷ 5

Variance = 2

Step 4: Calculate Standard Deviation

Standard Deviation = √2

Standard Deviation ≈ 1.41 GB

Interpretation

For this example, we consider:

- **Less than 3 GB = Low Standard Deviation → Consistent**
- **3 GB or more = High Standard Deviation → Varied**

Since **1.41 GB is less than 3 GB**, the mobile data usage is:

Consistent

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3. Comparison of Coding Test and Music Quiz

Coding Test Marks

60, 65, 70, 75, 80

Mean = **70**

Marks	Deviation	Squared Deviation
60	-10	100
65	-5	25

70	0	0
75	5	25
80	10	100

Sum = **250**

Variance

250 ÷ 5 = **50**

Standard Deviation

√50 ≈ **7.07**

Music Quiz Marks

40, 55, 70, 85, 100

Mean = **70**

Marks	Deviation	Squared Deviation
40	-30	900
55	-15	225
70	0	0
85	15	225
100	30	900

Sum = **2,250**

Variance

2,250 ÷ 5 = **450**

Standard Deviation

√450 ≈ **21.21**

Comparison

Test	Variance	Standard Deviation
Coding Test	50	7.07
Music Quiz	450	21.21

Conclusion

The **Music Quiz has more variation** because its variance (**450**) and standard deviation (**21.21**) are much higher than the Coding Test.

The Coding Test marks are more **consistent**, while the Music Quiz marks are more **spread out**.

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4. Standard Deviation for Special Cases

Case 1: Empty Data

If there are no observations, there is no variation to measure.

Standard Deviation = 0

Case 2: Only One Value

Suppose the data contains only one value:

50

Mean = **50**

Deviation = 50 − 50 = **0**

Therefore:

Variance = 0

Standard Deviation = √0 = 0

Summary

Situation	Standard Deviation
Empty data	0
One value	0

Conclusion

Variance and standard deviation are useful measures for understanding **how much data varies from its average**. A lower standard deviation means the data is more consistent, while a higher standard deviation means the data is more spread out.